Ensemble Sparse Kronecker Product Decomposition (E-CSKPD) to Reduce and Interpret High-Dimensional Tensors

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May 15, 2025

Abstract

This paper introduces generalizations and an ensemble method for nonlinear Sparse Kronecker Product Decomposition (SKPD) and its connection to convolutional neural networks (CNNs). In its entirety this method is referred to as the Ensemble Sparse Kronecker Product Decomposition (E-CSKPD) and we propose SKPD as a structured and interpretable method for analyzing high-dimensional image data with sparse signals. The nonlinear SKPD integrates statistical modeling with deep learning techniques, enabling enhanced predictive performance and efficient region detection in matrix and tensor data. We provide theoretical guarantees for coefficient estimation and region detection, along with simulation studies and a real brain MRI data analysis to illustrate its utility.

Keywords: Sparse Kronecker product, Convolutional neural network, Image regression, High-dimensional data, Tensor decomposition.

1 Introduction

Image regression and tensor decomposition have become central tools in high-dimensional data analysis. Traditional methods, such as canonical polyadic decomposition and Tucker decomposition, suffer from interpretability and computational challenges in large-scale datasets. To address these limitations, we propose the Sparse Kronecker Product Decomposition (SKPD), which

*The authors gratefully acknowledge the contributions of collaborators and funding agencies.

combines sparse modeling with Kronecker structure to enable region detection and efficient representation. In this paper, we extend SKPD to the nonlinear SKPD framework and establish its connections to convolutional neural networks (CNNs), a powerful tool in machine learning.

The rest of the paper is organized as follows: In Section 2, we introduce the one-term SKPD for matrix and tensor images, along with a path-following algorithm to solve one-term SKPD. Section 2.3 generalizes the method to to include generalizations to handle non-continuous terms for both the predictor and regressor variables. Section 3 describes tuning parameter selection and modifications extending SKPD into E-CSKPD and the enhancements to the path finding algorithm for the calculation of sparse matrices. Section 4 proposes the extension of nonlinearity within the SKPD, discusses its connections with CNNs, and the handling of ensemble models within the framework of SKPD. Section 5 provides theoretical guarantees on coefficient estimation and region detection. Simulation studies are presented in Section 6, and real brain MRI data analysis is discussed in Section 7.

2 Generalization of SKPD

2.1 Matrix Image Model

Consider the regression problem for 2D grayscale image data:

$$y_i = \langle X_i, C \rangle + \varepsilon_i, \quad i = 1, \dots, n,$$

where $y_i \in \mathbb{R}$ is the outcome, $X_i \in \mathbb{R}^{D_1 \times D_2}$ is the image data, $C \in \mathbb{R}^{D_1 \times D_2}$ is the unknown coefficient matrix, and ε_i are i.i.d. noises. The coefficient matrix C is modeled using a one-term Kronecker product decomposition:

$$C = A \otimes B, \quad \|A\|_F = 1,$$

where $A \in \mathbb{R}^{p_1 \times p_2}$ and $B \in \mathbb{R}^{d_1 \times d_2}$ are unknown matrices. The sparsity of A is enforced:

$$\|A\|_0 \leq s,$$

for some unknown sparsity level s. This forms the one-term Sparse Kronecker Product Decomposition (SKPD).

2.2 Tensor Image Model

The matrix SKPD model can be extended to higher-order tensors. For a three-order tensor, the SKPD model is:

$$y_i = \langle X_i, C \rangle + \varepsilon_i, \quad C = A \otimes B,$$

where C is a tensor with sparse A and structured B. The tensor SKPD framework enables efficient analysis of color or volumetric images.

2.3 Further Generalizations

CSKPD allows for additional tuning parameters beyond those of SKPD. These parameters are the canonical link function, g, and two different matrices to be convolved with the image data, K and Λ. Both K and Λ can be represented as a convolution of several matrices reflecting the overall transformation of the image matrix, X.

This equation is generalized as follows:

$$g(\mu) = \langle K * \Lambda * X_i, \Lambda^{-1} * C \rangle + \varepsilon_i, \quad i = 1, \ldots, n,$$

The architecture to convolve and deconvolve the tensors, X and C can be seen by the following architecture.

K and Λ can be chosen by best practices of image processing and signal detection in machine learning methods. For example, if the purpose is to de-noise a tensor standard kernels can be utilized to remove the random noise with K. Interpolation techniques and other kernels can be used to modify the tensor to conform with the desired effect. Those values can be convolved to form Λ. It is important to note that the transformations must be linear and invertible for use within CSKPD. This is an important property of reproducing the convolutions.

The separation of K and Λ allows for several useful techniques. It allows for operations that would yield information loss (dimension reduction through principal component analysis, de-noising by removing frequencies from the frequency domain, etc.) without it impacting the regions for detection. Thus convolutions that would impact the regions (interpolation for tensor reduction, kernels for edge detection and movement, etc.) can be retained for C. A special case of CSKPD is where $g(\mu) = \mu$, $K = I$, and $\Lambda = I$ returns the same results in the standard SKPD.

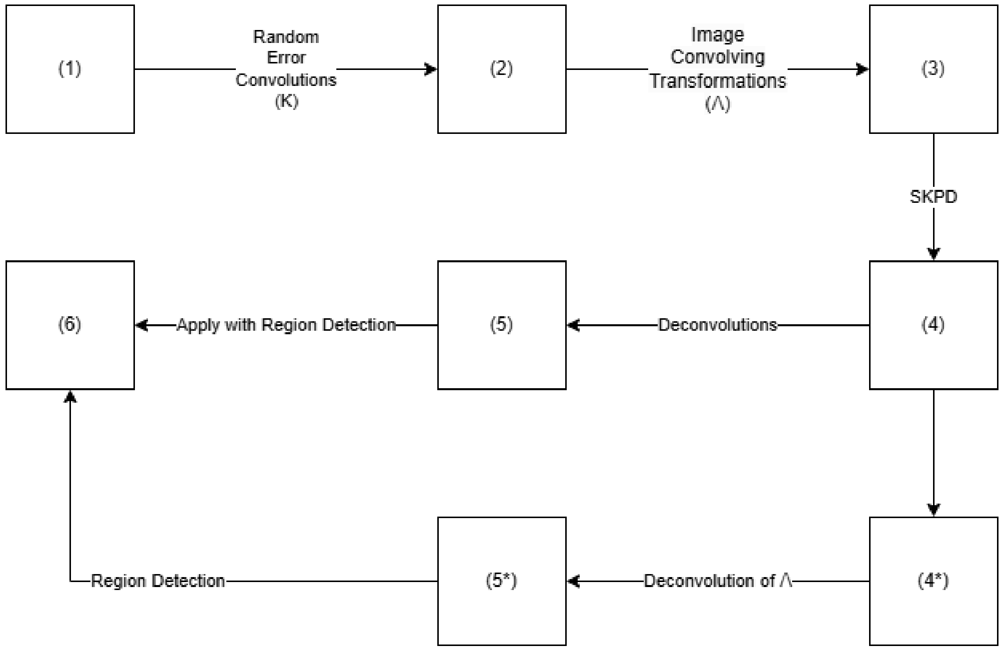

Figure 1: This summarizes the process of creating and recreating the tensor and region detection by using convolutions in the CSKPD design.

3 Modification of the Tuning Parameters

3.1 Updated MBIC

In the paper by Zak-Szatkowska, M., & Bogdan, M. (2011) [6], the authors propose a modified version of the Bayesian Information Criterion (MBIC), adapted from the MBIC model from Wang (2009) [4] specifically designed for sparse Generalized Linear Models (GLMs). This adaptation aims to include a model selection criterion that accounts for sparsity in general models. The formal modification suggested in their paper, published in *Computational Statistics and Data Analysis*, is given by:

$$\text{MBIC} = -2\log(L) + k\log(n) + 2k\log\left(\frac{m}{c} - 1\right)$$

where $c = 4$ is recommended as the standard choice.

Despite the implementation of cross-validation, there remains a susceptibility to overfitting. By utilizing the MBIC estimate as a basis for model selection, it is ensured that the chosen model not only adheres to the principles of parsimony but also effectively addresses the need for sparsity, thereby enhancing the accuracy and reliability of model predictions.

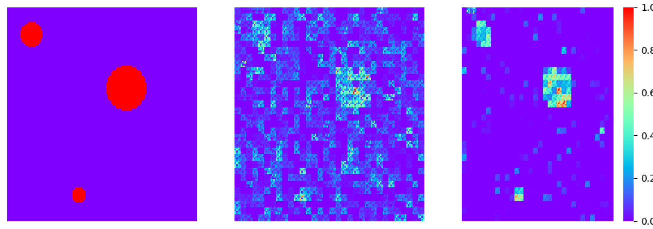

Figure 2: From left to right, true image, optimized for MSE, optimized for MBIC.

3.2 Enhancing the Path Finding Algorithm

SKPD provided an initial framework for determining the A and B matrices, whose Kroenecker product results in the matrix C. Since there is no closed-form solution for the optimal values of matrices or tensors A and B to minimize the least squares loss from **??**, a path finding algorithm is the most practical way to solve for A and B. However, there are some limitations to the path finding algorithm proposed by the SKPD method.

The first is a modification to the path finding algorithm was made to preserve the stability of matrix B and the sparsity in matrix A. SKPD provided an alternating value for λ. This causes issues with the convergence of A and B as A is not constrained by the same regularization parameter. It is noted that in Theorems 1 and 2 from the SKPD method [5] that $\lambda^{(t)} \propto^{(t)}$. Thus the formula for $\lambda(t)$ can be removed in favor of a grid search for λ_A and λ_B. This helps stabalize B without impacting any consistency or convergence constraints of the algorithm.

Additionally, in SKPD it is only proven that the criterion holds for a path finding algorithm for A and B and only says that this can be further generalized for additional parameters by regressing on the coefficients first and then performing SKPD on the residuals. The issue with that is it overfits the first model. This leads to overfitting to the training data and results in an SKPD model that is non-convergent for A and B as the underlying data is no longer directly reflective of the signal. To overcome this an additional parameter to the path finding algorithm is introduced, ζ. While it does take additional iterations to suffice the convergent criteria it is shown empirically to converge on data where the signal is retained. Additional analysis will be performed to prove this.

Algorithm 1 Alternating Minimization for the R-term SKPD with both ℓ_1 and ℓ_2 norms

Input: y_i and X_i, $i = 1, \dots, n$

Initialization: $\hat{A}^{(0)}$ is taken as the top left singular vector of $\sum_{i=1}^{n} X_i y_i$; with $\hat{X}_i = \mathcal{R}(X_i)$ for matrix image or $\hat{X}_i = \mathcal{R}(X_i)$ for tensor image, $\zeta^{(0)} = \vec{0}$, $z^{(0)} = \vec{0}$ and $(t) = 1, \dots, T$.

for $t \in \{0, 1, 2, \dots, T-1\}$ **do**

 $\hat{B}^{(t+1)} \leftarrow \min_B \frac{1}{2n} \sum_{i=1}^{n} (y_i - z_i^{(t)} - [\mathrm{vec}(B)]^\top \mathrm{vec}(\hat{X}_i^\top \hat{A}^{(t)}))^2 + \lambda_B \|\mathrm{vec}(B)\|_2$

 $\hat{A}^{(t+1)} \leftarrow \min_A \frac{1}{2n} \sum_{i=1}^{n} (y_i - z_i^{(t)} - [\mathrm{vec}(A)]^\top \mathrm{vec}(\hat{X}_i^\top \hat{B}^{(t+1)})^2 + \lambda_A \|\mathrm{vec}(A)\|_1$

 Orthogonalization: $\hat{A}^{(t+1)} \leftarrow \hat{A}^{(t+1)} (\hat{A}^{(t+1)\top} \hat{A}^{(t+1)})^{-1/2}$

 $\hat{C}^{(t+1)} \leftarrow \hat{A}^{(t+1)} \otimes \hat{B}^{(t+1)}$

 $\hat{\zeta}^{(t+1)} \leftarrow \min_\zeta \frac{1}{2n} \sum_{i=1}^{n} (y_i - \langle X_i, \hat{C}^{(t+1)} \rangle - Z_i^\top \zeta) + \lambda_\zeta \|\zeta\|_2$

 $z^{(t+1)} = Z^\top \zeta^{(t+1)}$

end for

return $\hat{A}^{(T)}$, $\hat{B}^{(T)}$, $\hat{C}^{(T)}$, $\hat{\zeta}^{(T)}$

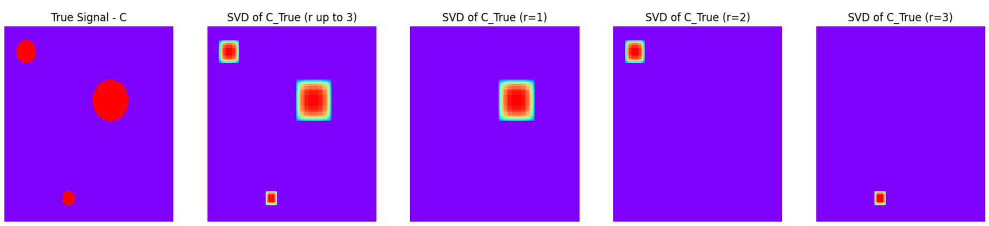

Figure 3: Principal Components of the true signal of circles.

3.3 Feature Extraction from $R = 1$

For any image where the data is properly segmented a signal can extract certain sections from an image. If the data is properly segmented, it extracts very well in the first r principal components. To demonstrate this we use the 'circles' signal provided by the initial SKPD method.

This is the goal of extracting the signal segments through SKPD, however when leveraging the $R > 1$ values from C_r, the segmentation of the signals are not properly identified.

It can be seen that despite the $\hat{C}$ signal being fairly well identified, each segmentation is not properly given. This is a key functionality of the SKPD method and a reason why the A_r matrix is initialized by using the first R principal components of the $\sum_i y_i X_i$. Furthermore the estimated signal for $\hat{C}$ provides similar signals when $R = 1$ and $R = 3$.

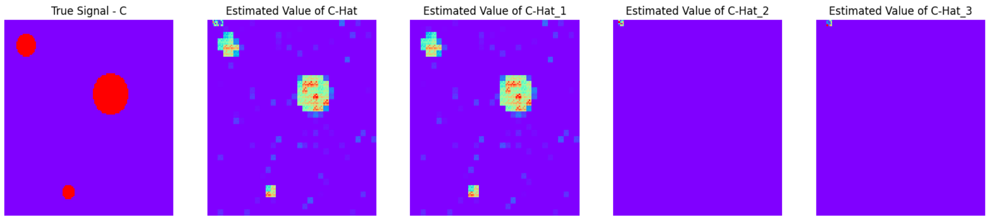

Figure 4: Each C_r from the true signal of circles.

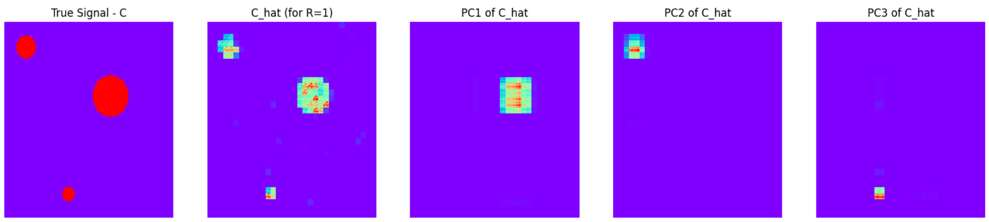

Figure 5: Principal Components of the $\hat{C}$ for $R = 1$ of the signal of circles.

Since $A_r \otimes B_r \in \mathbb{R}^{\gtrdot \times \ltimes} \quad \forall r \in R$ and $C = \sum_{r \in R} A_r \otimes B_r$, C is in the same vector space as each C_r. So it can be reproduced by any A, B. Thus for all models, each SKPD method can be simplified restricting to $R = 1$ to save compute time and then taking the first r principal components of C to extract the regions and use that in the predictions.

While the SKPD method focuses on a multi-region example, here we show that all of the benefits of SKPD can be utilized and extracted from $R = 1$.

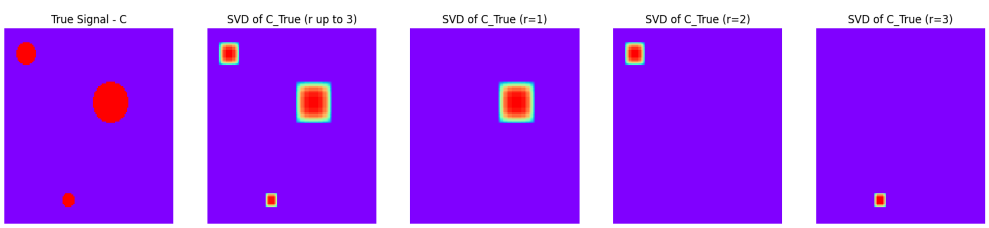

Figure 6: Principal Components of the true signal of circles.

4 E-CSKPD Tensor Regression

4.1 Rearrange Function Generalization

The function $\mathcal{R}$ is shown to act on partitions of the matrix/tensor. If we denote $P = [p, d]$ where p is the partition shape and d the sub-block dimensions, then:

$$\mathcal{R}(X, P) = \begin{bmatrix} \text{vec}(X_{1,1}^{d_1,d_2}) \\ \vdots \\ \text{vec}(X_{p_1,p_2}^{d_1,d_2}) \end{bmatrix}.$$

It follows that:

$$\mathcal{R}^{-1}(X, P) = \mathcal{R}(X, P^\top).$$

For the tensor case this can be further extended. This can be formulated by a flattening process can be represented as:

$$\bar{\mathcal{R}}(\mathcal{C}) = \left[\text{vec}(\mathcal{C}_{1,1,1}^{d_1,d_2,d_3}), \ldots, \text{vec}(\mathcal{C}_{p_1,p_2,p_3}^{d_1,d_2,d_3})\right]^\top.$$

Thus, $\mathcal{R}$ is shown to be an involution when $P^\top$ is used for the second application.

The text discusses a function called R that acts on partitions of matrices and tensors. The function takes two arguments: X, which is the matrix or tensor, and P, which represents the partition shape and sub-block dimensions. The output of the function is a flattened vector that represents the rearranged data.

The important aspect of this function is that it is shown to be an involution when used twice with the transpose of the second application. This means that applying the function twice in this way will result in the original matrix or tensor being returned. The authors provide a theoretical formulation of this process, which involves flattening the data and rearranging it into a vector using the vec function.

The text also mentions the use of convolutions to reduce the dimension size and ensembling the data to get similar results as the 3D tensor. This is because the rearrange function is invertible, meaning that the original data can be recovered from the flattened output. The specific details of these techniques are not provided, but their general importance in reducing the dimensionality of large datasets and improving model performance is highlighted.

4.2 Extension to Non-Linearity

Everything described thus far has been in the realm of frequentist statistics to ensure proper inference of the signal in the image, C. However, this method can be further expanded to include the non-linear behavior and produce a model that mimics the architecture of Convolutional Neural Networks.

The image convolving transformations can be extracted out of the hyperparameter and instead used as a model of models parameter. For each Λ within the model, the convolution can be performed on the image, with the constraint that the convolution is invertible. Then everything can be passed through the CSKPD model to determine the loss score for that model. A non-linear model can then be constructed, such as a tree-based model, can then be constructed using the loss of each model, resulting in the ensemble model that has a non-linear form.

Algorithm 2 Ensemble CSKPD: Meta-Learned Model Aggregation

Input: Training data $\{(X_i, y_i)\}_{i=1}^n$, optional covariates Z_i, base parameter sets $\{\theta_k\}_{k=1}^K$, common parameters θ_c and test data X_{test}, Z_{test}, initialize $\hat{Y}^*_{\text{test}} \in \mathbb{R}^{m \times K}$

Initialization: θ_c global parameters for all SKPD instances.

for $k = 1$ to K **do**

    $\theta'_k \leftarrow \theta_k \cup \theta_c$

    $f_k = \text{SKPD}(\{(X_i, y_i, Z_i)\}_{i=1}^n, \theta'_k)$

    $\hat{y}_i^{(k)} = f_k(X_i, Z_i)$ for $i = 1, \ldots, n$

end for

$\mathcal{M} \leftarrow [f_1, \ldots, f_K]$

$\hat{Y}_* = [\hat{y}^{(1)}, \ldots, \hat{y}^{(K)}]^\top \in \mathbb{R}^{n \times K}$

$w, b = \min_{w,b} \; \frac{1}{n} \sum_{i=1}^n \left(y_i - w^\top \hat{Y}_{*,i} - b \right)^2 + \lambda \|w\|_1 + \mu \|w\|_2^2$

for $k = 1$ to K **do**

    $\hat{y}_{\text{test}}^{(k)} = \mathcal{M}_k(X_{\text{test}}, Z_{\text{test}})$

end for

$\hat{Y}_{\text{test}} = [\hat{y}_{\text{test}}^{(1)}, \ldots, \hat{y}_{\text{test}}^{(K)}]^\top \in \mathbb{R}^{n \times K}$

$\hat{y}_{\text{final}} = \hat{Y}^*_{\text{test}} w + b$

return $\hat{y}_{\text{final}}$

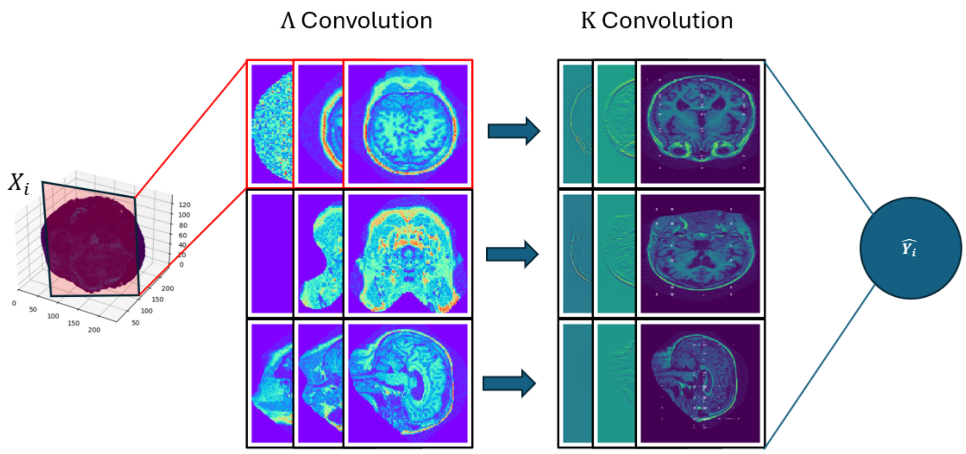

Figure 7: Ensemble of the different slices from the 3D tensor.

4.3 E-CSKPD Architecture

The nonlinear SKPD shares many similarities with convolutional neural networks (CNNs). Consider a CNN with one convolutional layer and one fully connected layer for predicting a scalar outcome from a $D_1 \times D_2$ image. Let R filters of size $d_1 \times d_2$ convolve with the input image X using strides (s_1, s_2) to produce feature maps of dimension:

$$\left\lfloor \frac{D_1 - d_1 + 1}{s_1} \right\rfloor \times \left\lfloor \frac{D_2 - d_2 + 1}{s_2} \right\rfloor .$$

The filters introduce sparsity and nonlinearity, similar to the role of A in SKPD. The nonlinear SKPD extends this analogy by incorporating activation functions, such as ReLU, to model complex interactions.

5 Theoretical Guarantees

5.1 Estimation Consistency of Generalized SKPD

In all cases, SKPD can be generalized to handle incoming data. It is assumed that all responses are obtained by independent samples which results in predictions of a response variable that has a distribution from the exponential family.

From these assumptions there exists a convolution K that when applied on the samples, X it forms a linear relationship with the response variable: $\eta = (K * X)\beta$. Also there exists a link function, g such that the expected value of the response variable $\mathbb{E}[Y] = \mu$ to the linear predictor: $g(\mu) = \eta$. These are the conditions of a generalized linear model. [2] This allows all data to be passed through as a generalized model and all of the conditions of SKPD hold.

From [5], Theorems 4 provides the following upper bounds of estimation consistency for the multi-term SKPD:

$$\frac{\left\|\sum_{r=1}^{R} \hat{A}^{(t)} \otimes \hat{B}^{(t)} - C\right\|_F}{\|C\|_F} \leq (\sqrt{R} + \kappa_2)\sqrt{\frac{\sum_{r=1}^{R} \|\hat{B}^{(t)} - B\|_F^2}{\sum_{r=1}^{R} \|B\|_F^2}} + \nu_2.$$

This upper bound simplifies to the following since r does not exist in the upper bound, and is defined for all values of R.

$$(\sqrt{R} + \kappa_2)\frac{\|\hat{B}^{(t)} - B\|_F}{\|B\|_F} + \nu_2 = (\sqrt{R} + \kappa_2)\sqrt{\frac{\sum_{r=1}^{R} \|\hat{b}_r - b_r\|_2}{\sum_{r=1}^{R} \|b_r\|_2}} + \nu_2$$

From equation 60 of [5]:

$$\leq (\sqrt{R} + \kappa_2)\sqrt{\frac{\sum_{r=1}^{R}(\kappa_1\kappa_2\mu_0 + \frac{\kappa_1\nu_2+\nu_1}{1-\kappa_1\kappa_2})\|b_r\|_2}{\sum_{r=1}^{R} \|b_r\|_2}} + \nu_2 = (\sqrt{R} + \kappa_2)c + \nu_2$$

Thus,

$$\frac{\left\|\sum_{r=1}^{R} \hat{A}^{(t)} \otimes \hat{B}^{(t)} - C\right\|_F}{\|C\|_F} \leq (\sqrt{R} + \kappa_2)c + \nu_2$$

meaning the upper bound can be defined by constants with the only parameter being the number of regions R. And since $(1 + \kappa_2) < (\sqrt{r} + \kappa_2) \quad \forall r > 1$, R can be set such that $R = 1$ and the estimation consistency holds and can be constrained in a smaller upper bound than that of $R > 1$.

Additionally, since $A_r \otimes B_r \in \mathbb{R}^{m \times n}, \quad \forall r \in R$ and $C = \sum_{r \in R} A_r \otimes B_r$, C is in the same vector space as each C_r. So it can be reproduced by any A, B. Thus for all models, each SKPD method can be simplified restricting to $R = 1$ to save compute time and then taking the first r principal components of C to extract the regions and use that in the predictions.

5.2 Estimation Consistency of E-CSKPD

5.3 Computational Complexity of E-CSKPD

Gradient boosted decision trees have a convergence rate of $O(\frac{1}{b+\lambda})$ where b is the number of boosted trees and λ is the learning rate. [3].

The convergence rate for SKPD is $\sqrt{\frac{n+sd_1d_2\log(p_1p_2)}{n}}$ from Theorem 2. [5]

6 Simulations

7 Real Data Analysis

7.1 OASIS-1

The initial data set used for the proposed Cyclic-Voxel SKPD method is the OASIS-1 dataset. This dataset contains brain scans of 436 patients and this information is stored as a medical imaging file. [1] This image file is created by taking numerous scans across three coordinate directions of the body: coronal, horizontal, and sagittal view of the human brain. These slices detect the signal as the MRI machine scans each segment of the brain. This information can be converted into a 3D tensor for data analysis.

Figure 8 shows how the information is captured within the tensor. It is important to note a couple key components of this data. The first is that it is absent of a lot of signal data on the outer perimeter of the scan. These regions will not provide any additive accuracy to the model. Additionally, there is a lot of noise in between each slice due to the process of how the MRI captures the signal data. This is also something to incorporate in the model.

The OASIS-1 dataset also provides extensive metadata for each subject, including demographic details such as age, sex, handedness, education level, and socio-economic status. Additionally, it includes clinical information such as the Mini-Mental State Examination (MMSE) scores, Clinical Dementia Rating (CDR), estimated total intracranial volume (eTIV). Normalized whole brain volumes (nWBV) is also a provided value based on a estimate of the calculated region surrounding the brain. The key for this data is determining the CDR score associated with each patient from the MRI scan, this is a cognitive test associated with whether or not a patient can be diagnosed with Alzheimer's. A score of less 0.5 or above is a key indicator.

After doing some initial analysis there appears to be a strong correlation between age and sex with the MMSE score. In addition, the nWBV appears to have strong correlation with the MMSE score, and because it is a simple and deterministic metric from the calculation of the scans, this is a value that can be used in regression. A naive approach of taking a regression model from the CDR score and corresponding covariates and an AUC score were 0.6067 with age, sex, and nWBV as covariates.

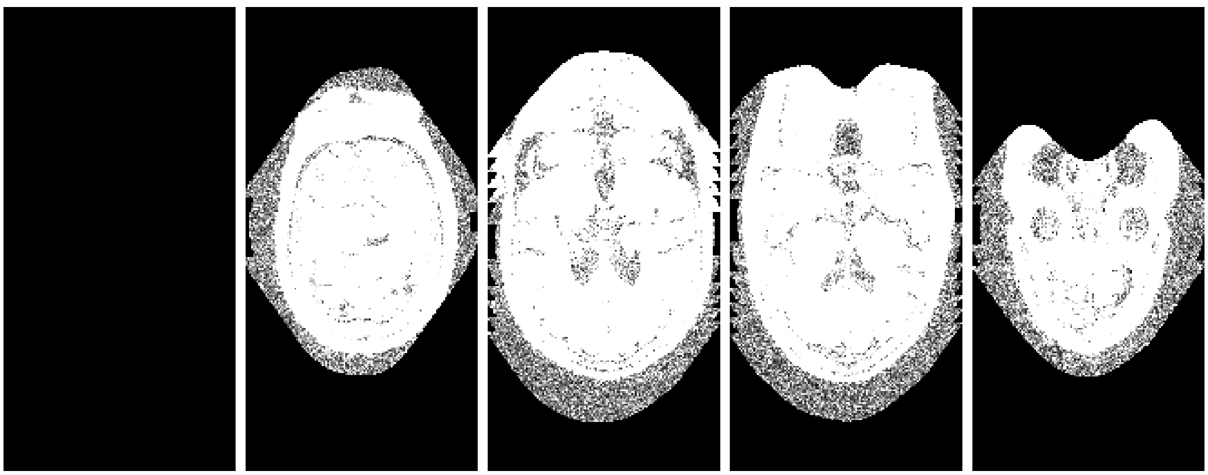

Figure 8: Snapshots of slices of the coronal view of the brain, showing the progression of the scan from anterior to posterior, highlighting the noise of the scans.

Table 1: 5-fold Cross Validation AUC Scores for Alzheimer's Detection

Model	**Performance Metrics**	
	Average	**Best**
Best traversal slice	(0.6123)	(0.7208)
Best coronal slice	(0.6988)	(0.8953)
Best sagital slice	(0.8027)	(0.9048)
Best traversal slice w/ kernel	(0.6627)	(0.8056)
Best coronal slice w/ kernel	(0.6730)	(0.7696)
Best sagital slice w/ kernel	(0.6808)	(0.7927)
Ensemble Model	(0.6800)	(0.9000)

7.2 ADNI

Similar to the OASIS-1 data, we wish to obtain whether a patient has Alzheimer's by examining their MRI brain scans. For this, the ADNI1: Complete 1Yr 1.5T data set is used. This data set provides several different brain scans. It too collects age and sex, but does not output brain volume.

Each patient is categorized as AD indicating Alzheimer's Disease, MCI indicating Mild Cognitive Impairment, or CN indicating Cognitively Normal. These are classified as a binary response for cognitively normal similar to that of the CDR in the OASIS-1 data set.

The data extraction is performed in a similar manner as well. However it is important to make sure each observation is coming from a similar machine, thus only those with a (256, 256, 166) image size is extracted. Extracts slices from 3D volumes (transverse, coronal, sagittal). These images are then Resizes images to a standard size (96x96) for consistent processing and median filtering and cropping non-zero regions to improve image quality.

The text describes a methodology used in machine learning for evaluating the performance of models on classification tasks. This approach involves implementing a 5-fold cross-validation strategy, which means that the data is divided into five parts and the model is trained and evaluated five times, each time using a different part of the data as the test set and the remaining four parts as the training set.

The methodology also involves defining an ensemble model using multiple feature extraction methods, which means combining the results of several different approaches to feature selection to improve the performance of the model. This is done by comparing the results of standard vs gradient-based filtering.

Furthermore, the model is trained with varying parameters, including different kernels and normalization techniques. Kernels are mathematical functions that are used to transform the input data into a form that is more suitable for machine learning algorithms, while normalization techniques are used to adjust the scale of the data so that all features are on a similar scale.

Finally, the performance of the model is evaluated using AUC (Area Under the Curve) for classification tasks, which is a metric that measures the ability of a classifier not to err on the ambiguous instances. The predictions made by the model are compared against ground truth labels to assess its performance. Below are the results.

Table 2: 5-fold Cross Validation AUC Scores for Alzheimer's Detection on ADNI data

Model	**Performance Metrics**	
	Average	**Best**
Best traversal slice	(0.5030)	(0.5618)
Best coronal slice	(0.4925)	(0.5667)
Best sagital slice	(0.5108)	(0.5373)
Best traversal slice w/ kernel	(0.5148)	(0.5714)
Best coronal slice w/ kernel	(0.5154)	(0.6250)
Best sagital slice w/ kernel	(0.5361)	(0.6037)
Ensemble Model	(0.5516)	(0.6444)

8 Conclusion

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